Personalizing RAG for Enterprise Analytics

Santosh Reddy Muthyala





Grounded answers still miss the person asking

- Enterprise analytics is now conversational, and RAG grounds answers in governed information
- Grounding alone does not make an answer useful to its reader

Three users, three needs

E-commerce checkout: why did conversion decline?

Conversion = completed checkout sessions / sessions that started checkout.

Data engineer

- Metric definition
- Executable logic
- Data lineage
- Query assumptions
- Data validation
- Fact verification

Business analyst

- Size of change
- Segment breakdown
- Comparable time windows
- Evidence and hypotheses
- Validation approach
- Clear explanation

Product manager

- Concise KPI summary
- Link to dashboard
- Affected journey
- Next steps
- Decision checkpoint

Overall	Mobile	Desktop
5.0% to 4.5%	6.0% to 5.0%	4.0% to 4.0%

Hypothetical. 10,000 checkout starts per device in each of two complete weeks.

Personalize the evidence and the expression, never the facts

Stays fixed

Shared, auditable analytical truth

Adapts

Retrieval emphasis, detail level, format and follow-ups

Presentation preferences are kept apart from access rights, definitions and facts.

Engineer output

Checkout conversion query

Engineer

SQL

```
SELECT week, COALESCE(device,'Overall') AS device,
  COUNT(*) AS starts,
  SUM(completed_flag) AS purchases,
  AVG(1.0 * completed_flag) AS conversion
FROM eligible_checkout_sessions
GROUP BY GROUPING SETS ((week, device), (week));
```

Metric: completed sessions / eligible starts

Grain: one checkout session

Query result

Device	Prior	Last
Mobile	6.0%	5.0%
Desktop	4.0%	4.0%
Overall	5.0%	4.5%

Checks pending

Definition Lineage Assumptions

Illustrative UI. Hypothetical data. Payment-error causation remains unconfirmed.

Analyst output

Checkout conversion insights

Analyst

Conversion	Change (percentage points)	Mobile share (both weeks)
4.5%	-0.5	50%

Checkout conversion



Mobile explains the full decline

Device mix is unchanged.

Payment errors: 2.0% to 3.5%

Explore providers and versions

Illustrative UI. Hypothetical data. Payment-error causation remains unconfirmed.

Product manager output

Checkout decision brief

Product manager

Mobile checkout needs investigation

-10%

relative conversion

5.0% to 4.5%

100

fewer completed sessions

Checkout volume unchanged

Suggested owner

Checkout + Payments

Next step

Review errors and recent releases.

Open checkout dashboard

Decision: fix or rollback after validation

Illustrative UI. Hypothetical data. Payment-error causation remains unconfirmed.

Personalization is a controlled decision at every step

01

Identity and permissions

What may be seen

02

Profile

Likely goals and style

03

Retrieval

Eligible evidence

04

Ranking

What is useful now

05

Generation

Organized response

06

Feedback

Did it help?

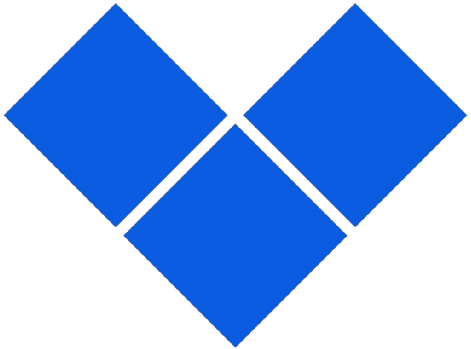
Three components, with explicit contracts

User Profile Store

Centralized storage of user data and preferences.

Personalized Generation Layer

Generates tailored outputs based on retrieved context.



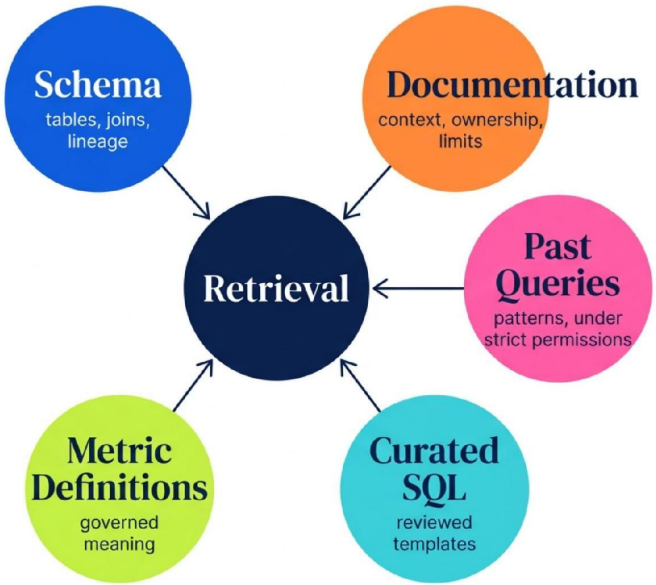
Multi-Index Retrieval System

Aggregates and retrieves signals from multiple indexes.

A profile holds hints, not rules

- Keep the minimum attributes that improve the task: role, format, domain, tools, dataset familiarity
- Record provenance: declared, inferred, imported or application-supplied
- The current request overrides history; users can view, correct and reset
- Familiarity with a restricted dataset never grants access

Five indices answer different questions



Authority and entitlement come before similarity

01

Entitlement checks

Run before evidence reaches a model context

02

Structured filters

Environment, business unit, metric and time window

Conflicts are surfaced, not hidden, to avoid confirmation bias.

Ranking prioritizes useful, authoritative evidence

1 Establish eligibility

Source permissions

Business unit and time window

Metric definitions and authority

2 Prioritize eligible evidence

Relevance to the question

Fit for the current task

Freshness where it matters

Popular evidence may still be wrong. Conflicting evidence stays visible.

Checkout example: a popular query cannot override the metric definition.

Personalized generation turns evidence into a useful answer

The prompt contract specifies how the model should use the evidence and format the answer.

Inputs

Current request

Editable profile hints

Permitted evidence

→

Response contract

Structure and depth

Requested format

Sources and uncertainty

→

Tailored answer

Engineer: query and checks

Analyst: segment breakdown

PM: impact and next action

Shared guardrails

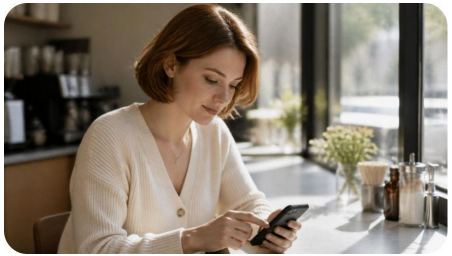
Preserve facts

Cite evidence

Expose uncertainty

Version and test prompt templates. Treat retrieved text as data.

One question, three answers: why did conversion decline?



Same evidence

Users can switch modes

Engineer

- SQL-first
- Canonical definition
- Ingestion-lag and duplicate checks

Analyst

- Segment narrative
- Window alignment
- Correlation versus causation

Product manager

- Trend, affected journeys
- Dashboard link
- Visible uncertainty

Build around real workflows, with a phased rollout

01

Unify source metadata and access controls

02

Introduce task-aware retrieval

03

Add user-editable response preferences

04

Evaluate bounded profile inference

Evaluate retrieval, answers, drift and safety

Retrieval

- Relevant, authoritative, permitted, timely.

Answers

- Factual support, citations, depth, format fit.

Drift

- Reduce personalization when profiles go stale.

Safety

- Prompt injection, query-history exposure, cross-tenant leakage.

Start with a trustworthy evidence foundation

- Inventory metrics, schemas, docs and reusable queries
- Attach ownership and validity metadata
- Enforce authorization before context assembly
- Keep human review for consequential interpretations and actions

Align explanation with intent while facts, permissions and accountability stay stable



Thank you

Santosh Reddy Muthyala

